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Analytics, Intelligence, and Technology Division

A Methodological Overview of Seismic Analysis for Nuclear Event Detection

25 May 2022

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A Methodological Overview of Seismic Analysis for Nuclear Event Detection

Introduction

Underground explosions generate potentially detectable signatures, including energy waves that travel through the Earth's subsurface (i.e., seismic waves), low-frequency sound waves (i.e., infrasound and hydroacoustic waves), and radioactive gases and/or particles that might leak from the test cavity (if the event was nuclear). There can also be intelligence indicators of a test, such as observations of modified patterns of life and activity at a suspected test site. If all of these detectable signatures and intelligence indicators are present and self-consistent, then analysts have high confidence in classifying a signature-generating event as an explosion. However, because only partial information about an event is likely to be available, determining whether an event was natural (e.g., an earthquake or landslide) or manmade (e.g., a chemical or nuclear explosion) is much more challenging. This primer describes how one category of event signatures—seismic signatures—can augment event analyses. While universities and government organizations have generated detailed technical descriptions of seismic analytic techniques,[1,2,3,4,5,6,7,8,9,10] we seek to translate seismic event analysis for a broad, non-technical audience.

When the geologic conditions[a] near an event are well-characterized, seismic data can be used to calculate critical information, such as event location and depth, with relatively high accuracy. Moreover, specific features within seismic datasets can help determine whether an event was an explosion. However, a key challenge in seismic analysis is that geologic site conditions are often poorly characterized, complicating the ability to discern the true nature of the event. To overcome this challenge, geologists answer a series of questions (discussed in section 1) to guide seismic event analysis and determine the most probable nature of an event. As more information is gathered during each analytic step, confidence grows regarding the nature of the event. Section 2 addresses uncertainties in seismic analysis and the vital nature of high-fidelity geologic data for accurate seismic event analysis.

[a] Conditions include (1) rock type(s) at the event site and along the path between the event and seismic stations, (2) geologic material properties (e.g., density) at the event site, and (3) the location and orientation of geologic structures (faults, folds, fractures) at and near the event site.

Seismic Event Analysis Overview

When a natural or manmade underground event occurs, energy travels away from the source in all directions through the Earth's subsurface in the form of seismic waves. These seismic waves can travel for thousands of kilometers and are eventually detected by a **seismometer**, which is mounted inside an instrument called a **seismograph** (see appendix A). Seismographs are located worldwide via numerous seismic monitoring networks (see appendix B). Seismographs record each event as a **seismogram** (figure 1), which is a recording of ground movement at an instrument's location. The horizontal axis is time, and the vertical axis is ground displacement; higher peaks on the vertical axis indicate more ground motion, which correlates to a larger event. When there is no earthquake, the seismogram records background noise that corresponds to local disturbances (e.g., vehicle traffic, construction, industrial processes).

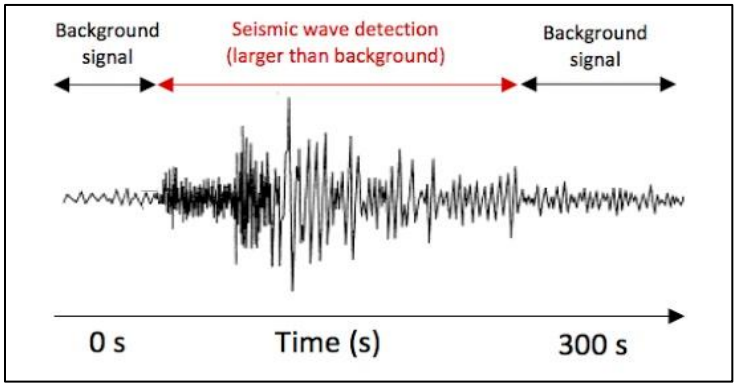


Figure 1. Notional seismogram recorded by a seismograph. All monitoring stations have a small amount of background seismic signal due to everyday activities. Earthquakes and explosions alter this seismic signal by producing larger-amplitude seismic waves.

To determine whether an event recorded as a seismogram is an earthquake or an explosion,[b] geologists pose a series of questions to examine the event. Figure 2 summarizes these questions in a step-by-step process using notional seismograms from an earthquake (top figure, middle column) and an underground nuclear test (top figure, right column) and demonstrates how the answers to each of these questions differ depending on the nature of the event.

The first three questions determine event location, size, and depth. While none of these questions alone can diagnose whether an event was natural or manmade, the answers to them provide critical information to guide additional analysis. Likewise, none of the subsequent three questions, which assess the probable source of a seismic detection, can by themselves diagnose the source of an event, but consistency among the answers increases confidence in event classification. If questions 1 through 6 indicate that an event is probably a manmade explosion, event size is used to estimate event yield (question 7).

[b] While other natural events (landslides, avalanches, slope failures) and manmade events (construction, hydrofracturing, football games) can generate seismic waves, this primer focuses on the difference between natural earthquakes and manmade explosions.

Using seismogram data recorded at monitoring stations, the following seven questions (left-hand column) are used to analyze an event and determine the probable source of an event (natural or manmade):

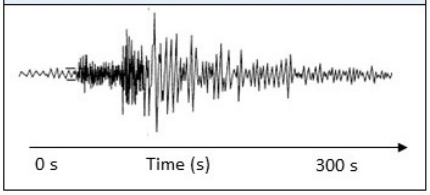
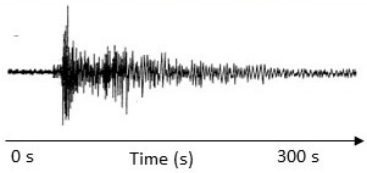
	Natural (Earthquake)	Manmade (Nuclear)
An underground event generates seismic waves. These seismic waves are recorded at monitoring stations in the form of seismograms.	**Notional earthquake seismogram:** 	**Notional nuclear test seismogram:** 
1. Where did the event occur?	**If the event was an earthquake...** ...it probably occurred in a historically seismic region	**If the event was a nuclear test...** ...it occurred in a historically seismic region OR in a location where earthquakes are uncommon
2. How big was the event?	**If the event was an earthquake...** ...it could exhibit a wide range of magnitudes	**If the event was a nuclear test...** ...the event magnitude will be relatively small (< 7.0)
3. How deep was the event?	**If the event was an earthquake...** ...the event depth will be > 5 km	**If the event was a nuclear test...** ...the event depth will be < 2 km
4. What is the general appearance of seismic waveforms?	**If the event was an earthquake...** ...first and second arrivals and surface wave signals are prominent	**If the event was a nuclear test...** ...the first arrival signal is large with little to no second or surface wave signatures
5. What do the first arrival waveforms look like?	**If the event was an earthquake...** ...the first arrival waves are a mix of compressive and dilational waves	**If the event was a nuclear test...** ...all first-arrival waves appear compressive
6. Are surface waves present?	**If the event was an earthquake...** ...surface waves are present with relatively large amplitude and duration	**If the event was a nuclear test...** ...surface waves are either not present or present with minimal amplitude and duration
7. What is the approximate yield?	**If the event was a nuclear test, additional analyses can provide more information:** Event yield is calculated using event magnitude (question 2). Event yield greater than 500 tons TNT equivalent is a probable nuclear explosion; event yield less than 500 tons requires more detailed analyses.	

Figure 2. Analytical questions designed to determine whether a seismic event was natural or manmade.

Question #1 Where did the event occur?

Seismograms from at least three monitoring stations are required to calculate the location of a seismic event. Geologists use the difference in seismic wave arrival time[c] at each monitoring station to triangulate approximate event location (see appendix C).

- **Natural or manmade?** A single event located in an area typically devoid of earthquakes and mining activities could signal a manmade event of concern. In contrast, a series of ton-scale events in an area of known commercial mining operation might be less concerning (or at least merit additional analysis to confirm event context).
- **Analytical uncertainties:** Every seismic event location has an associated error ellipse due to uncertainties about geologic structure and parameters both at the event site and along the path between the approximate event location and detecting seismic stations. Error ellipses are typically on the order of tens of kilometers but can be reduced with additional station detections.

Question #2 How big was the event?

Event size is reported as **magnitude**, a dimensionless number that represents the relative size of an event on a scale from 1 (very small event) to 10 (very large event). Thousands of small earthquakes (magnitude <= 2.0) occur every year, but only approximately 20 larger earthquakes (magnitude > 7.0) occur annually (figure 3). Multiple methods can be used to calculate event magnitude using various subtypes of seismic waves (see appendix D). Magnitude is a key parameter for calculating yield if subsequent analysis indicates a manmade event.

- **Natural or manmade?** It is very unlikely that an event larger than $m_b = 7$ would be manmade;[11,d] events smaller than $m_b = 7$ could be natural or manmade in origin.
- **Analytical uncertainties:** Magnitude can vary broadly depending on the angle between the seismometer and the event due to the radiation pattern of the source. Additionally, different magnitude calculation methods yield different values and are not always suitable for larger (magnitude 6+) events.

[c] Primary waves are compressional waves that apply a force in the direction of propagation. Primary waves transmit their energy easily, travel quickly (~5–8 km/s), and are typically the first seismic waves to be registered at a seismic monitoring station. Secondary waves are transverse waves that cause material to shake at right angles to the direction of motion. Secondary waves transmit their energy less efficiently, travel slower than primary waves (~3–5 km/s), and therefore arrive at seismic monitoring stations after primary waves.

[d] The largest US underground nuclear test, the 5-megaton CANNIKIN test, generated a magnitude of 7.02.[11] We judge that it is very unlikely—*within the current geopolitical landscape*—that a proliferant country would risk the political fallout from detonating such a large device.

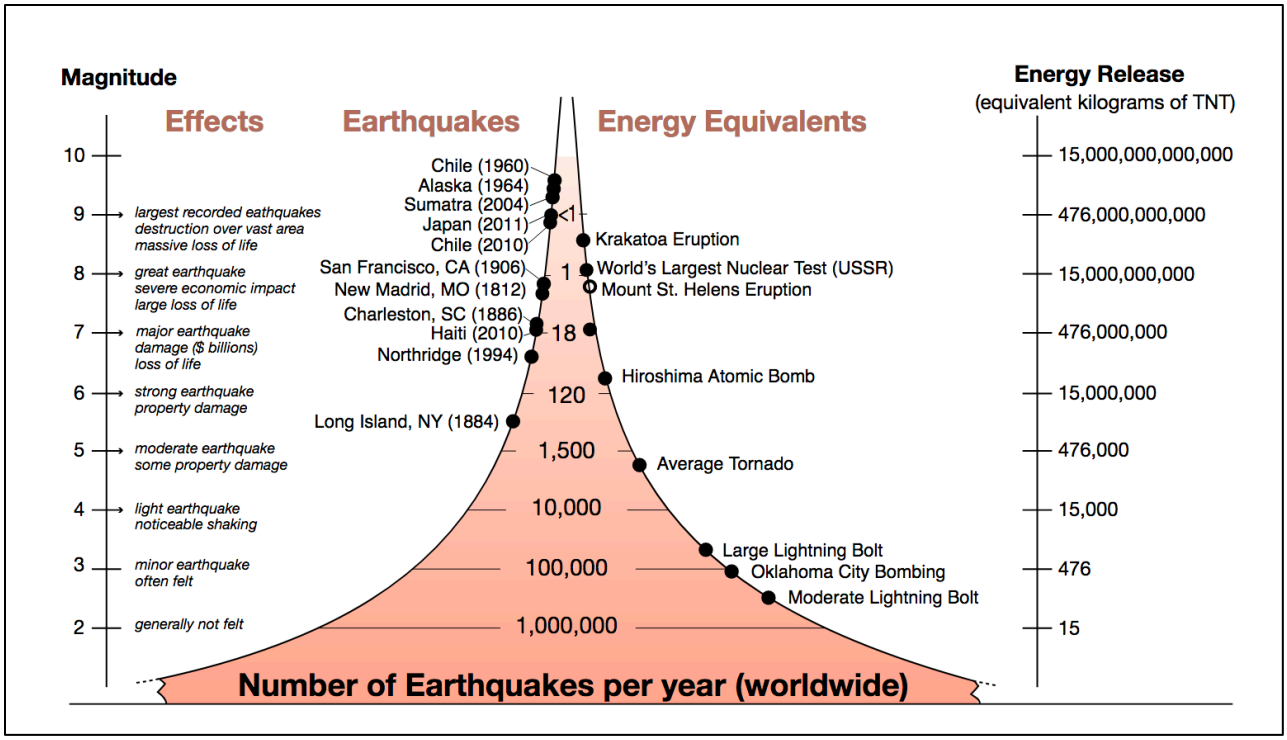


Figure 3. Earthquake magnitude and associated energy release. Figure from US Geological Survey.

Question #3: How deep was the event?

Seismic waves travel in all directions—including upward—away from the event location and eventually intersect with, and then reflect away from, the Earth's surface. These reflected seismic waves are known as "pP waves" or **depth phases**. The time delay at monitoring stations between the arrival of primary seismic waves and these depth phases is proportional to the depth of the event (figure 4).

- **Natural or manmade?** Manmade events such as nuclear tests and mining blasts typically occur within the upper 2 km of the Earth's surface. The United States' largest underground nuclear test, the 5-megaton CANNIKIN test, had a depth of burial of 1.9 km, while most underground nuclear tests were detonated between 200 m and 700 m depth of burial.[12] A handful of the world's deepest gold mines—which can generate seismic signals from blasting—reach depths near 4 km. By contrast, natural earthquakes typically occur at depths of at least 5 km and up to tens to hundreds of kilometers below the Earth's surface.
- **Analytical uncertainties:** Particularly in mountainous regions, abrupt changes in topography can cause the travel time of reflected waves to be highly variable, even within a small area. Therefore, the depth of very shallow seismic events is one of the most difficult parameters to calculate.

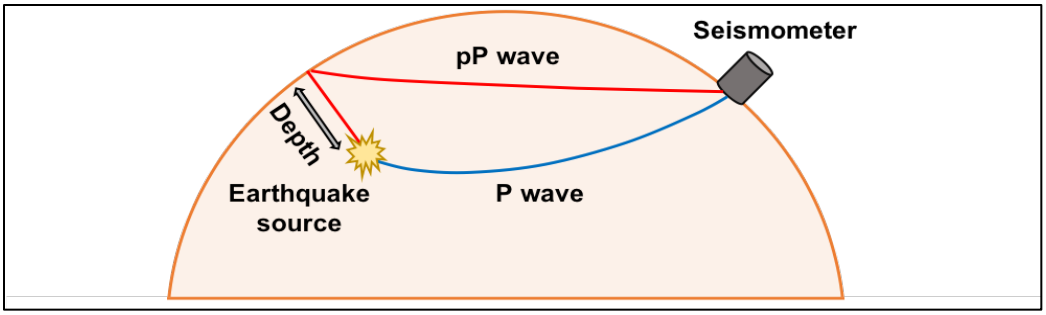


Figure 4. Notional illustration of primary waves (P, blue line) and depth phases (pP, red line). The pP wave travels a longer distance than the primary wave and therefore arrives at the monitoring station later.

Question #4: What is the general appearance of the seismic waveforms?

Primary and secondary waves—collectively known as **body waves**—travel through the Earth's inner layers and are detected by sensors far away. The amplitudes of faster-moving primary waves and slower-moving secondary waves depend on whether the event was an earthquake or a manmade event, so the ratio of primary wave amplitudes to secondary wave amplitudes is often used to determine whether an event was natural or manmade.

- **Natural or manmade?** Due to the generation of large amounts of shear stress, earthquakes generate weak primary waves and strong secondary waves, resulting in pronounced secondary wave arrival in a seismogram. Conversely, due to minor amounts of shear stress, explosions generate strong primary waves and weak secondary waves. The seismograms in figure 5 demonstrate the different arrival appearances and amplitudes of primary and secondary seismic waves depending on event type.
- **Analytical uncertainties:** All monitoring stations experience a small amount of background seismic signal due to everyday activities such as vehicle traffic, industrial processes, and construction projects. This background signal can obfuscate the transition from background seismic noise to event seismic signal (particularly for very small events), and the appearance of seismic waveforms can be difficult to decouple from background noise.

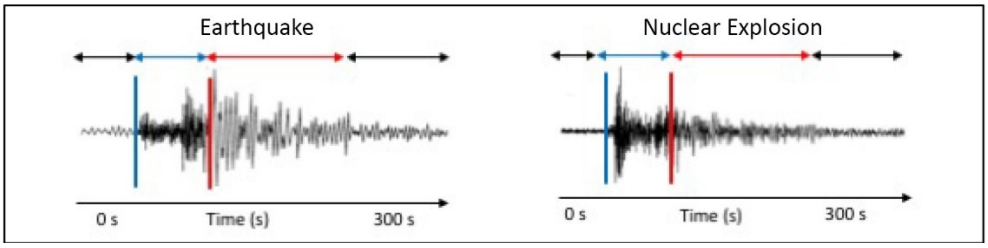


Figure 5. The notional seismogram at left is from an earthquake with relatively weak primary wave arrivals (blue) and stronger secondary wave arrivals (red). The notional seismogram at right is from a nuclear explosion with relatively strong primary wave arrivals followed by weaker secondary wave arrivals.

Question #5: What do the first arrival waveforms look like?

The characteristics of the first arrival waveforms—i.e., up (compression) or down (dilation)—can be used to distinguish an earthquake from a manmade explosion. Compressive waveforms result from rocks squeezing together; dilational waveforms result from rocks being pulled apart (see appendix E).

- **Natural or manmade?** During an earthquake, some rocks are compressed while others are dilated relative to the fault location, so seismic monitoring networks will record a mix of compressive and dilational waves. Seismograms recorded after a natural event will record a mix of both "up" and "down" signals. However, because manmade explosions apply relatively uniform pressure to the cavity walls and cause seismic waveforms to radiate uniformly in all directions, seismic waveforms from an underground explosion will only be compressional, and seismograms recorded after a manmade event will record only "up" signals.
- **Analytical uncertainties:** Similar to the analytical uncertainties associated with the previous question, the transition from background seismic noise to event seismic signal can be obfuscated due to low signal-to-noise ratio, and the appearance of the first arrival seismic waveforms can be difficult to decouple from background noise.

Question #6: Are surface waves present?

Explosions represent a point source in space and time whereas earthquakes result from faults shifting over tens of kilometers; the more localized and instantaneous nature of explosions tends to reduce or eliminate the presence of surface waves compared to earthquakes.

- **Natural or manmade?** The presence of substantive surface waves in a seismogram indicates an earthquake. While explosive events can create surface waves, these waves are of minimal amplitude and duration and are difficult to discern in seismograms, particularly in seismometers located very far from a seismic event.
- **Analytical uncertainties:** Low signal-to-noise ratios—particularly in the case of very small explosions—will often conceal the presence of surface waves.

Question #7: What is the approximate event yield?

If questions 1–6 indicate that the event is likely manmade, event yield is calculated using event magnitude (see appendix F). Depending on the event magnitude and corresponding event yield, the event might be of great interest to the Intelligence Community. For example, the September 2016 North Korean test generated an event magnitude of 5.3,[13] which is equivalent to approximately 3–11 kilotons, assuming a fully coupled explosion and standard scaled depth of burial, or 16–22 kilotons if the test was overburied.[14] On a smaller scale, an event magnitude of 2.8 is equivalent to a yield of approximately 20 tons.[15,e] If the magnitude indicates a yield greater than 500 tons, the event was probably a nuclear rather than conventional (i.e., chemical) explosion. An event less than 500 tons might also indicate a nuclear explosion; however, the use of hundreds of tons of chemical explosives in mining is not without precedent, so one would have to verify that such use of chemical explosives did not coincide with the recorded event. Additionally, large explosions typically detonate in serial or "ripple-fired" explosions.[16] One effect of ripple-firing is that the seismic signals resemble an earthquake more than an explosion.[17]

[e] Assuming a fully coupled shot in hard rock. See tonebox on p. 11 for more information about fully coupled and decoupled explosions.

The margins of error for event yield are often large and will vary depending on (1) how well the geological conditions for the event area are understood, (2) where seismic instruments are positioned relative to the event of interest, and (3) possible evasion techniques (see tonebox, p. 11).

Additional Uncertainties in Seismic Event Analysis

Using seismic data alone, it is challenging to determine whether a manmade event was a conventional or nuclear explosion.[18,f] In 1993, DOE detonated a ~1-kiloton conventional explosive device near historic nuclear testing sites to compare the seismic signals from chemical and nuclear explosions; unfortunately, diagnostic data revealed no unique seismic signatures that can be leveraged to discriminate between chemical and nuclear events.[19,20] However, seismic detections still augment event assessments by providing contextual information, such as event magnitude and yield.

The primary challenge in using seismic data to characterize underground explosions and determine event yields is uncertainty about geologic structure and parameters at the event site and along the path between the event and seismic stations.[21] High-fidelity data on rock type(s), geologic properties (e.g., density), and the location and orientation of geologic structures (e.g., faults, folds, fractures) are critical for accurate seismic analyses because these parameters control how seismic waves travel through the subsurface. Characterization of local and regional geology identifies locations that focus, scatter, or otherwise disrupt seismic waves; if these effects are identified near recording stations, detailed seismic analysis can be modified for improved site-specific analysis.

Site characterization ideally requires physical access to allow ground-based measurements, seismic calibration tests near the event site, and installation of geodetic monuments.[g] However, even without physical access, many tools and techniques can be employed to better understand geologic data.

- ***Geomorphometric analyses*** integrate terrain analysis and pattern recognition techniques to improve surface geologic characterization.[22] These analyses rely on studies that have demonstrated a relationship between surface roughness and geology.[23,24] For example, variable rock types and geologic structures result in quantifiably different stream drainage patterns due to variable weathering patterns. An open-source, space-borne digital elevation model can be used to extract drainage patterns over a broad area and connect these drainage patterns to underlying geologic information. A geomorphometric approach extracts terrain characteristics that approximate underlying geology in the absence of ground-based data.
- ***Synthetic aperture radar (SAR)*** can be leveraged to quantify possible surface displacement. Integrating space-borne elevation data with seismic records provides additional information about possible deformation following an underground event.[25]
- ***Geologic mapping via satellite imagery*** provides basic information about large-scale faults and other broad geologic features (figure 6). While vegetation often makes it difficult to

[f] Currently, an explosion can only be classified as nuclear once radioactivity is detected.

[g] Geodetic monuments, also known as survey monuments or survey benchmarks, are permanent fixtures on the Earth's surface that provide latitude, longitude, and orthometric height and provide a spatial location reference of a point or area.

perform geologic mapping via satellite imagery, this technique can be useful in desert regions where vegetation is minimal and outcrops/exposure are easily identified.

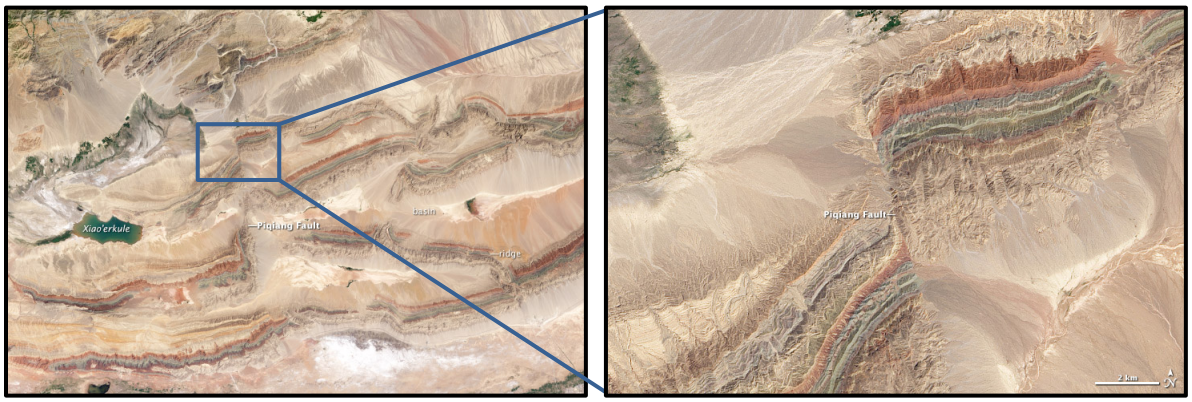


Figure 6. Images of the Keping Shan thrust belt from the Landsat 8 satellite. Note the pronounced offset of geologic units due to the Piqiang Fault, particularly in the inset image (right).

Though they cannot provide complete information for accurate seismic event analysis, these tools can reduce uncertainty in seismic analyses by providing contextual information that can be used to approximate the subsurface structure when comprehensive geologic data is unavailable.

Lastly, sparse seismic station coverage and the possibility that an explosion was deliberately masked or muted (see tonebox, p. 11) pose challenges for seismic analysis. In the absence of comprehensive geologic data, analysts are forced to make informed assumptions, approximations, or outright best guesses on variables that can be critical to the accuracy of a seismic analysis.

Masking a Manmade Explosion

Decoupling: In standard underground explosive testing, a chamber just large enough for the test device is carved out of the earth. When the device explodes, energy in the form of seismic waves is transmitted to the surrounding rock and then travels to a seismic station where the event is measured. In a decoupled test, experimenters construct a cavity much larger than required for event containment. When the device explodes, much of the energy is transmitted into increased temperature and pressure within the larger cavity, thereby significantly reducing the seismic waves emanating from the test. Decoupling increases the chance that an event will go undetected or that its size will be underestimated. The United States has a history of successful decoupling, including the STERLING, SALMON, and MILL YARD tests.

Mine masking: Operations and associated chemical explosions from mines near an underground testing facility could be used to mask a small nuclear explosion by creating additional seismic waves that confuse or obfuscate the event signal. However, mining operations tend to use ripple-fired explosions (i.e., detonated in a rapid sequence or chain) that in aggregate do not generate seismic waves as efficiently as a single explosion of the same total yield. Hence, a single large explosion amongst several smaller explosions could still create a signature.

Testing during an earthquake swarm: An earthquake swarm is the occurrence of many earthquakes in a local area within a relatively short period of time (typically days to months). The swarm creates a steady stream of seismic waves that have the potential to overload monitoring stations. The detonation of a nuclear device during a swarm would make it difficult for the monitoring community to deconflict the signals and recognize the difference between natural and manmade events, so an underground explosion might go unnoticed. However, because it is impossible to predict when an earthquake swarm will occur, experimenters would have to maintain constant test readiness while awaiting a swarm.

Appendix A: Seismographs, Seismometers, and Seismograms

A **seismograph** is an instrument that measures seismic waves. Early models of seismographs have a **seismometer** mounted inside which is essentially a mass mounted on a spring (figure 7). Seismographs are securely mounted to the Earth's surface; when the ground shakes, the entire unit shakes *except* for the seismometer. As the seismograph shakes around the mass, the recording device on the mass records the relative motion between itself and the rest of the instrument, thus recording the ground motion. Newer seismometers work electromagnetically: a large, permanent magnet is mounted inside while the outside case contains bundles of fine wire. Movements of the case relative to the internal magnet generate electrical signals in the wire coil.

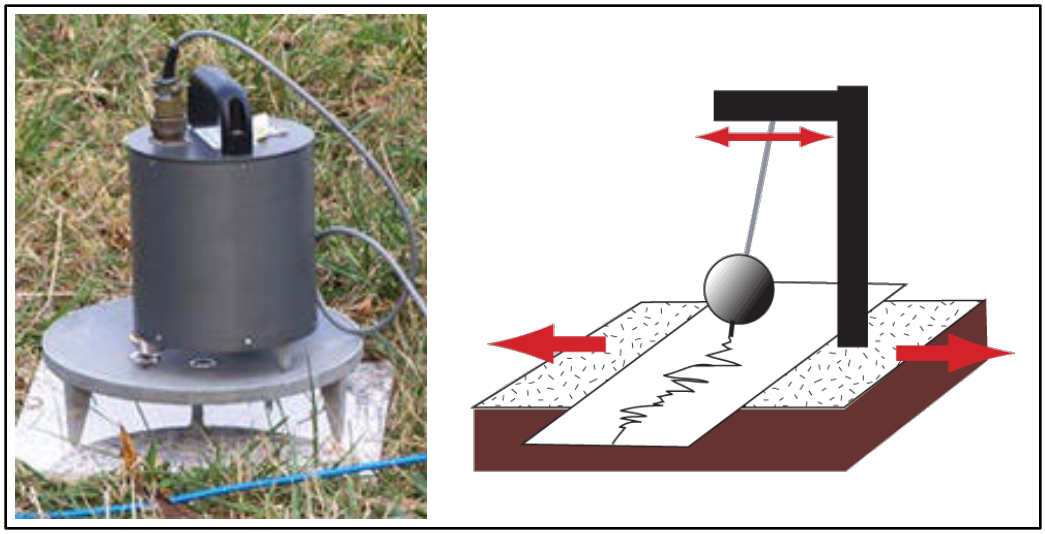


Figure 7. Seismograph (left) and seismometer (right). Images from US Geological Survey.

Seismograms are the recording of the ground shaking at the location of the instrument (figure 8). The horizontal axis is time, and the vertical axis is ground displacement; higher peaks on the vertical axis indicate more ground motion, which correlates to a larger event. When there is no earthquake, the seismogram is a nearly straight line except for very minor recordings that correspond to local disturbances (e.g., traffic, construction).

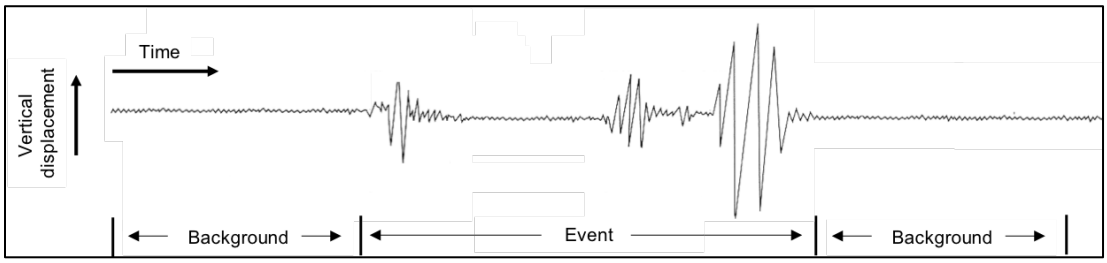


Figure 8. Notional seismogram. Seismograms includes a seismic event (large peaks in the center) and background measurements from local disturbances.

Appendix B: Seismic Monitoring Networks

Seismographs are installed around the globe and can be installed as standalone units or as part of a seismic monitoring network. There are numerous seismic monitoring networks in use today. The Preparatory Commission for the Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO PrepCom) promotes a worldwide verification regime that is designed to detect any nuclear explosion on Earth. The CTBTO PrepCom accomplishes this in part through the International Monitoring System (IMS), which includes seismic, hydroacoustic, infrasound, and radionuclide sensors deployed around the world. The installation of seismographs is ongoing, with 170 seismic stations located or planned to be installed in 76 countries. More than half of the primary seismic stations will be small seismic arrays with groups of 9–25 sensors arranged over a large spatial area. The use of arrays improves (1) signal-to-noise ratio, (2) azimuthal estimation of incoming seismic signals (i.e., the direction that the seismic waveform has travelled from), and (3) estimation of seismic wave speed.

In addition to the CTBTO IMS, a number of single seismic stations and regional monitoring networks across all continents provide seismic data through the Global Seismographic Network (GSN). GSN is a partnership between the US Geological Survey (USGS), the National Science Foundation (NSF), the Incorporated Research Institutions for Seismology (IRIS), and other global partners that aim for near-uniform distribution of seismic stations worldwide. GSN data are first archived by IRIS and then become part of the USGS' National Earthquake Information Center (NEIC) and Earthquake Hazards Program Comprehensive Earthquake Catalog (ComCat). These stations and regional monitoring networks are primarily installed to record strong-motion events that have significant human impacts, such as large earthquakes in major population centers.

Lastly, the International Seismological Centre jointly maintains the International Registry of Seismograph Stations, and the Consortium of Organizations for Strong Motion Observation Systems (COSMOS) is a non-profit organization that provides access to ground motions recorded anywhere in the world through its Virtual Data Center. COSMOS' primary objective is global dissemination of strong motion data.

Appendix C: Calculating Event Location

Underground events generate three types of seismic waves (figure 9):

- ***Primary waves*** travel in the direction of wave propagation in a "push-pull" pattern by compressing and expanding the surrounding material. Average P-wave velocity is approximately 5–8 km/s (the actual value depends on the geologic unit through which the wave is traveling). Primary waves can travel through solids, liquids, and gasses.
- ***Secondary waves*** travel perpendicular to the direction of wave propagation in an "up-down" pattern. Average S-wave velocity is approximately 3–5 km/s (the actual value depends on the geologic unit through which the wave is traveling). Secondary waves can travel through solids but cannot travel through liquids or gases.
- ***Surface waves*** travel along the surface at velocities slower than 3 km/s. Surface waves are almost entirely responsible for damage during an earthquake.

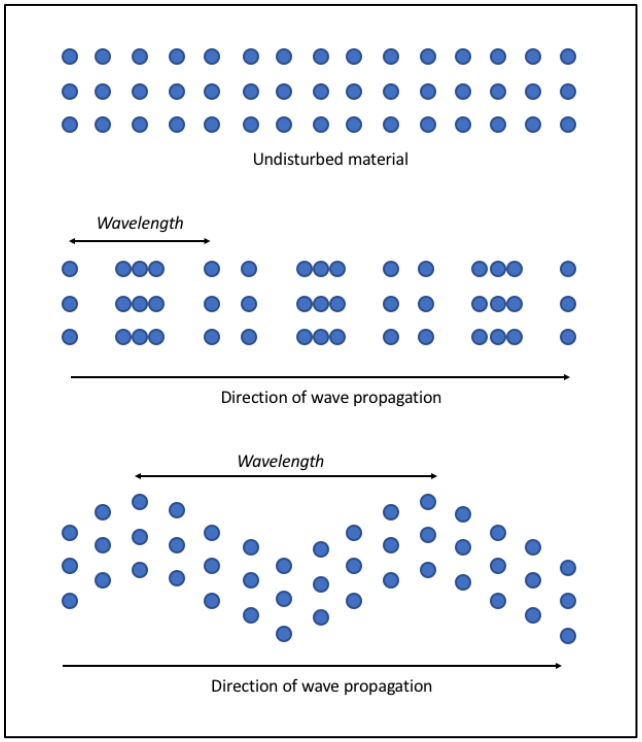


Figure 9. Schematic of undisturbed material (top), primary waves (center), and secondary waves (bottom). Primary waves travel in a push-pull fashion and secondary waves in an up-down fashion.

The difference in velocity values between P- and S-waves means that they arrive at seismographs at different times. This difference in arrival time is used to determine event location. Figures 10 and 11 present a notional example of how primary and secondary wave arrival times are used to determine event location. In this example, approximate travel times are plotted for primary and secondary seismic waves (figure 10). The analyst starts by determining the difference in arrival time between the P and S

waves for each of the three seismograms and then uses the difference to determine how far away each seismic monitoring station is from the event.

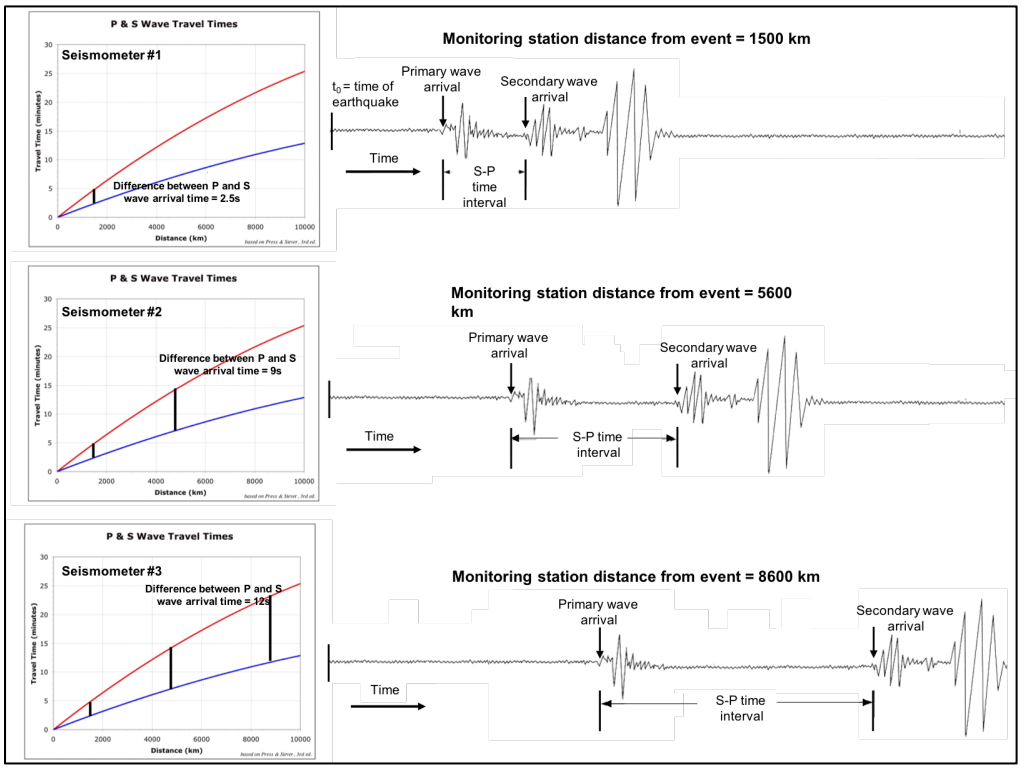


Figure 10. The left-hand charts plot the difference between P and S wave arrival times for three seismometers. The right-hand plots show the notional seismogram recorded at each seismometer.

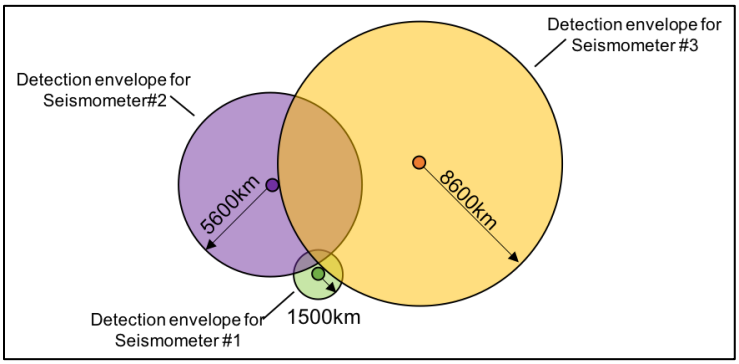


Figure 11. Detection envelopes of each seismometer from figure 10. The area where all three error ellipses intersect indicates approximate event location.

Appendix D: Calculating Event Magnitude

Several scales exist for calculating event size or **magnitude,** including (1) local magnitude (M_L), also known as the Richter scale, (2) surface-wave magnitude (M_s), (3) body-wave magnitude (M_b), and (4) moment magnitude (M_W). Because all magnitude calculations use a logarithmic scale, an increase of one unit of magnitude on a magnitude scale is equivalent to an increase of 10 times the amplitude recorded by a seismograph and approximately 30 times the energy.

Local magnitude (M_L) or **Richter Scale** is one of the earliest algorithms for quantifying the strength of an earthquake. M_L calculates event size based on the amplitude of the largest seismic wave that is recorded by a seismograph and is independent of seismic wave type (i.e., primary, secondary, or surface wave). The original formula for calculating M_L is

$$M_L = \log(A_{max}) - \log(A_0)$$

where A_{max} is the largest or *zero-to-peak* amplitude on a particular brand of seismograph (Wood-Anderson) and $\log(A_0)$ is a calibration value based on the distance of the seismograph from the event location.[25] Local magnitude calculations can be refined by using $\log(A_0)$ values that have been calibrated using regional seismometer stations.[25]

Other algorithms and methods have largely replaced the Richter scale, but this term is still commonly used as a catch-all name for earthquake magnitude.

Surface-wave magnitude (M_S) uses surface waves—also called Love and Rayleigh waves—to determine event magnitude and is calculated by measuring the zero-to-peak amplitude of the arriving surface waves. Shallow events typically generate larger surface waves compared to deep earthquakes; therefore, M_S is primarily only used to calculate magnitude of shallow earthquakes since it underestimates the size of deep earthquakes. The original formula for calculating M_S is

$$M_S = \log(A) + 1.656 \log(\Delta) + 1.818$$

where A is the horizontal component of the maximum ground displacement (in microns) due to surface waves with periods of 20 seconds and Δ is distance in radians.[25] However, the Prague formula is now the widely accepted formula for surface-wave magnitude:[25]

$$M_S = \log\left[\frac{A}{T}\right]_{max} + 1.66 \log(\Delta) + 3.3$$

where T is the period of the maximum surface-wave amplitude in seconds and the $(A/T)_{max}$ value is the maximum of all A/T values of the wave group.

Body-wave magnitude (M_b) uses body waves (i.e., primary and secondary waves) to determine event magnitude and is primarily used for deep earthquakes that do not typically generate large surface waves. The formula for calculating M_b is

$$M_b = \log\left[\frac{A}{T}\right] + q(\Delta, h)$$

where $q(\Delta, h)$ is an empirical correction depending on distance Δ and focal depth h.

Moment magnitude (M_w) measures the magnitude of an event using the **seismic moment** (M_O) rather than seismic wave amplitude. M_O is based on the area of fault rupture, average amount of slip, and the force required to overcome the friction sticking the rocks together that were offset by faulting. M_O is defined as

$$M_O = Da\mu$$

where D is the average fault displacement, a is the total area of the fault surface, and μ is the average rigidity of the rocks in the fault. M_O is used to calculate M_W using the following equation:

$$M_W = 2/3 \log M_O - 10.7$$

M_W is associated with earthquake source processes and eliminates the possibility of **magnitude saturation**, which occurs when earthquake magnitude calculations rely on peak amplitude, causing magnitude calculations to cluster around particular numerical thresholds. For this reason, M_W is the most reliable method for calculating the relative size of larger earthquakes.

Appendix E: Seismic Wave Propagation Mechanisms

Seismologists use seismograms to plot the "up" signals from body wave compressions and the "down" signals from body wave dilations using black-and-white diagrams called focal mechanism plots, or "beach ball" diagrams (figure 12). Focal mechanism plots are circular diagrams divided into regions that represent directions in which seismic waves travel following an event. These plots offer geologists the opportunity to evaluate fault orientation and areas of compression and dilation.

Focal mechanism plots appear completely white before an earthquake and then are shaded black in certain areas once seismometers register an "up" or "down" signal in a region. Earthquakes generate variable focal mechanism patterns in which there are some black and some white regions depending on whether each region experiences compression (black) or dilation (white) (figure 12).

By contrast, focal mechanism plots appear completely white before an explosion and then are shaded completely black following an explosion. This is because there is fairly uniform distribution of energy in all directions from an explosion—there is no direction away from the explosion that receives more energy than any other.

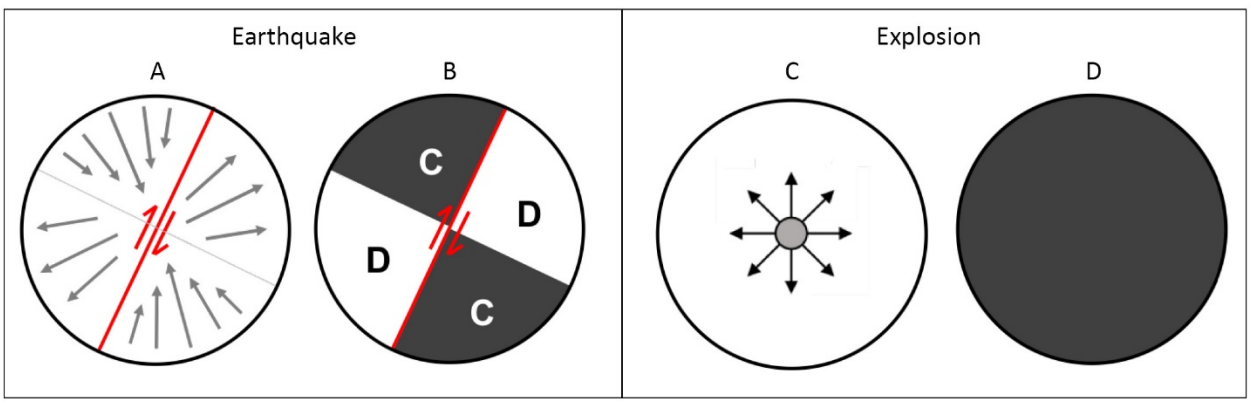


Figure 12. (A) When an earthquake occurs along a fault (red line), some regions experience compression (arrows pointing towards the fault) and other regions experience dilation (arrows pointing away from the fault). (B) The focal mechanism plot as a result of this earthquake, where some regions experience compression and other regions experience dilation. (C) When an explosion occurs, energy radiates away from the source in all directions. (D) The focal mechanism plot as a result of this explosion, where all regions experience compression.

Appendix F: Calculating Explosive Yield

Explosive yield is calculated from magnitude using the following equation:

$$M = A + B \log(Y) + \textit{bias correction}$$

where Y is yield in kilotons, *A* and *B* are constants that are dependent on local geology, and the *bias correction* is an adjustment to correct for how efficiently seismic waves travel.[25] While these constants have been calculated for well-characterized locations such as the Nevada National Security Site, there is a high degree of uncertainty in these values for many foreign test sites.

References

[1] Mueller, R.A., and Murphy, J.R. | 1971 | Seismic characteristics of underground nuclear detonations part I. Seismic spectrum scaling | *Bulletin of the Seismological Society of America* | Vol. 61 | pp. 1675–1692.

[2] Sipkin, S.A. | 1982 | Estimation of earthquake source parameters by the inversion of waveform data: Synthetic wave-forms: Physics of Earth and Planetary Interiors | Vol. 30 | pp. 242–259.

[3] Lay, T. | 1985 | Estimating explosion yield by analytical waveform comparison | *Geophysical Journal of the Royal Astronomical Society* | Vol. 82 (1) | pp. 1– 30 | https://doi.org/10.1111/j.1365-246X.1985.tb05126.x.

[4] US Congress Office of Technology Assessment | May 1988 | Seismic Verification of Nuclear Testing Treaties | OTA-ISC-361 | Washington D.C., U.S. Government Printing Office.

[5] Walter, W.R., and Priestley, K.F. | 1991 | High-frequency *P* wave spectra from explosions and earthquakes. In Explosion Source Phenomenology | *Geophysical Monograph* | Vol. 65 | pp. 219– 228 | Washington DC: American Geophysical Union | https://doi.org/10.1029/GM065p0219.

[6] Cronin, V. | 2004 | A Draft Primer on Focal Mechanism Solutions for Geologists | https://www.ocean.washington.edu/courses/oc410/reading/Focal_mechanism_primer.pdf.

[7] Taylor, N. | 25 June 2018 | Drums in the deep: Estimating test yields based on seismic signal | https://www.armscontrolwonk.com/archive/1205515/drums-in-the-deep-estimating-north-korean-nuclear-test-yields-based-on-seismic-signal/.

[8] Allen, R. | 2005 | Nuclear Yield | https://seismo.berkeley.edu/~rallen/research/nuke/yield.html.

[9] Comprehensive Nuclear-Test-Ban Treatment Organization (CTBTO) | 2020 | Seismic Monitoring | https://www.ctbto.org/verification-regime/monitoring-technologies-how-they-work/seismic-monitoring/.

[10] Karasözen, E., and Karasözen, B. | December 2020 | Earthquake location methods | *GEM – International Journal of Geomathematics* | Vol. 11 | https://doi.org/10.1007/s13137-020-00149-9.

[11] Willis, D. et al. | 1972 | Seismological and Related Effects of the CANNIKIN Underground Nuclear Explosion, Final Report | doi: 10.2172/4613144 | https://www.osti.gov/servlets/purl/4613144.

[12] FAS | 2000 | Underground Nuclear Weapons Testing | https://fas.org/nuke/guide/usa/nuclear/testing.htm.

[13] Gaber, H. et al. | 2017 | Seismological investigation of September 09 2016, North Korea underground nuclear test | *NRIAG Journal of Astronomy and Geophysics* | Vol. 6 | pp. 278–286 | doi: 10.1016/n.nrjag.2017.09.002.

[14] Zhao, L-F. et al. | 2017 | The 9 September 2016 North Korean Underground Nuclear Test | *Bulletin of the Seismological Society of America* | Vol. 107 (6) | pp. 3044–3051 | doi: 10.1785/0120160355.

[15] National Research Council | 2012 | The Comprehensive Nuclear Test Ban Treaty: Technical Issues for the United States | Washington, DC: The National Academies Press | doi: 10.17226/12849.

[16] Khalturin, V.I., Rautian T.G., and Richards P.G. | 1998 | The seismic signal strength of chemical explosions | *Bulletin of the Seismological Society of America* | Vol. 88 (6) | pp. 1511–1524.

[17] US Congress Office of Technology Assessment | May 1988 | Seismic Verification of Nuclear Testing Treaties | OTA-ISC-361 | Washington D.C., U.S. Government Printing Office.

[18] Richards, P., and Zavales, J. | 1990 | Seismic Discrimination of Nuclear Explosions | *Annual Review of Earth and Planetary Sciences* | Vol. 18 (1) | pp. 257–286.

[19] Denny, M. et al. | 1996 | Seismic Results from DOE's Non-Proliferation Experiment: A Comparison of Chemical and Nuclear Explosions | *Monitoring a Comprehensive Test Ban Treaty* | doi.org/10.1007/978-94-011-0419-7_20 | https://link.springer.com/chapter/10.1007/978-94-011-0419-7_20.

[20] Selby, N.D., Marshall, P.D., and Bowers, D. | 01 February 2012 | m_b:M_s Event Screening Revisited | *Bulletin of the Seismological Society of America* | Vol. 102 (1) | pp. 88–97 | doi:10.1785/0120100349.

[21] Rodi, W., and Toksöz, M.N. | October 2001 | Uncertainty analysis in seismic event location | *23rd Seismic Research Review: Worldwide Monitoring of Nuclear Explosions* | pp. 324–332 | https://www.researchgate.net/publication/235105925_Uncertainty_Analysis_in_Seismic_Event_Location.

[22] Coblentz, D., Pabian, F., and Prasad, L. | 2014 | Quantitative Geomorphometrics for Terrain Characterization | *International Journal of Geosciences* | Vol. 5 | pp. 247–266 | doi: 10.4236/ijg.2014.53026.

[23] Dellwig, L.F., and Moore, R.K. | 15 July 1966 | The Geological Value of Simultaneously Produced Like- and Cross-Polarized Radar Imagery | *Journal of Geophysical Research* | Vol. 71 | pp. 3597–3601 | https://doi.org/10.1029/JZ071i014p03597.

[24] Dellwig, L.F., Kirk, J.N., and Walters, R.L. | 15 October 1966 | The Potential of Low-Resolution Radar Imagery in Regional Geologic Studies | *Journal of Geophysical Research* | Vol. 71 (20) | pp. 4995–4998 | https://doi.org/10.1029/JZ071i020p04995.

[25] Wang, T. et al. | 13 July 2018 | The rise, collapse, and compaction of Mt. Mantap from the 3 September 2017 North Korean nuclear test | Science | Vol. 361 (6398) | pp. 166–170 | doi: 10.1126/science.aar7230.